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Image forming apparatus

Abstract

An image forming apparatus includes an image forming unit, a casing, a door, a connecting member, a guide member, and an operating portion. The operating portion moves, through an operation by a user when the door is at a first opened position, from a regulating position where the connecting member is regulated from moving from a first guide to a second guide, to an allowing position where the connecting member is allowed to move from the first guide to the second guide. When the operating portion is at the regulating position, the door is configured to be retained at the first opened position by engagement of the connecting member and the first guide. When the operating portion is at the allowing position, the door is configured to be retained at the second opened position by movement of the connecting member from the first guide to the second guide.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to image forming apparatuses adopting an electrophotographic system, such as a facsimile, a printer, or a multifunction machine having such functions.

Description of the Related Art

(2) Hitherto, in image forming apparatuses, a configuration in which a conveyance path is formed as an openable/closable cover is adopted to cope with jamming of sheets, hereinafter referred to as sheet jamming, in the conveyance path through which sheets are conveyed. In addition to removing the sheets that caused sheet jamming, the cover is opened and closed when attaching, detaching or replacing components and units, such as an intermediate transfer belt, a fixing unit, and a conveyance roller, that are arranged along the conveyance path. A configuration in which an opening angle of a cover is set to two stages, which are a small angle and a large angle, has been proposed (refer to Japanese Patent Application Laid-Open Publication No. 2014-21135). For example, since the opening angle for removing jammed sheets may be set small, the cover is opened to a small angle so as to reduce the processing space required for removing jammed sheets. Further, by switching the opening angle to a larger angle, attaching and detaching of components

can be performed conveniently.

(3) The above-mentioned configuration proposes a two-action system as a unit for switching the opening angles of the cover, wherein the cover is lifted to a point where the opening angles are switched, and while the cover is held in this state, a member supporting the cover is operated to switch angles. Actually, a guide groove that guides a guide shaft at a tip of a support arm disposed swingably on a casing is formed on the cover, and a small angle path, a large angle path, and a branching point are formed in the guide groove. Thereby, the operation for switching from the small angle to the large angle may be performed by an operation of lifting the support arm and switching the guide shaft from the small angle path to the large angle path. According to this configuration, when switching the opening angle, the support arm must be held by the user, such that the cover will not be opened with force unintentionally from the small angle to the large angle, and damaging of components caused by impact may be prevented.

(4) However, according to the image forming apparatus disclosed in Japanese Patent Application Laid-Open Publication No. 2014-21135, when opening the cover from the small angle to the large angle, there is a need to lift the support arm while guiding the guide shaft from the small angle path to the large angle path. According to this configuration, the load of the cover is received by the support arm of the cover, such that a large load is applied to the support arm, and this state is even more outstanding in a printer corresponding to large-size sheets or a high-speed apparatus in which the cover has a large size and has various functions. Therefore, in order to open the cover from the small angle to the large angle, such a heavy support arm must be lifted while performing the cover opening operation, such that a large operating force is required and the operability is deteriorated. Since the operability is not good, the operation of opening the cover to a large opening angle could only be performed by operators having special skills. Therefore, simplification of the operability was desired to allow general users to operate the cover.

(5) The present technique provides an image forming apparatus capable of simplifying the operability of opening the cover in two stages.

SUMMARY OF THE INVENTION

(6) According to a first aspect of the present invention, an image forming apparatus includes an image forming unit configured to form an image on a sheet, a casing accommodating the image forming unit, a door configured to be opened and closed with respect to the casing, and configured to be moved to a first opened position and a second opened position, the first opened position being a position where the door is inclined by a first angle with respect to a closed position, where the casing is closed and a sheet conveyance path is formed, to expose the sheet conveyance path, the second opened position being a position where the door is inclined by a second angle that is greater than the first angle with respect to the closed position to expose the sheet conveyance path, a connecting member configured to be interlocked with an opening and closing operation of the door, and connected to the casing and the door, a guide member provided on the door, the guide member including a first guide configured to guide the connecting member such that the door is positioned at the closed position and the first opened position respectively, and a second guide branched from the first guide and configured to guide the connecting member such that the door is positioned at the second opened position, and an operating portion configured to move, through an operation by a user in a state where the door is at the first opened position, from a regulating position where the connecting member is regulated from moving from the first guide to the second guide, to an allowing position where the connecting member is allowed to move from the first guide to the second guide. In a state where the operating portion is at the regulating position, the door is configured to be retained at the first opened position by engagement of the connecting member and the first guide. In a state where the operating portion is at the allowing position, the door is configured to be retained at the second opened position by movement of the connecting member from the first guide to the second guide.

(7) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cross-sectional view illustrating a general configuration of an image forming apparatus according to a first embodiment.
- (2) FIG. 2A is a front view of a cover in a state where the cover according to the first embodiment is positioned at a closed position.
- (3) FIG. 2B is a front view of a guide portion in a state where the cover according to the first embodiment is positioned at the closed position.
- (4) FIG. 3A is a front view of the cover in a state where the cover according to the first embodiment is positioned at a first opened position.
- (5) FIG. 3B is a front view of the guide portion in a state where the cover according to the first embodiment is positioned at the first opened position.
- (6) FIG. 4A is a front view of the cover in a state where the cover according to the first embodiment is positioned at a second opened position.
- (7) FIG. 4B is a front view of the guide portion in a state where the cover according to the first embodiment is positioned at the second opened position.
- (8) FIG. 5A is a view illustrating the guide portion viewed from one side of the guide portion in a state where the cover according to the first embodiment is positioned at the closed position.
- (9) FIG. 5B is a view illustrating the guide portion viewed from the other side of the guide portion in a state where the cover according to the first embodiment is positioned at the closed position.
- (10) FIG. 6A is a view illustrating the guide portion viewed from one side of the guide portion in a state where the cover according to the first embodiment is positioned at the first opened position.
- (11) FIG. 6B is a view illustrating the guide portion viewed from the other side of the guide portion in a state where the cover according to the first embodiment is positioned at the first opened position.
- (12) FIG. 7A is a view illustrating the guide portion viewed from one side of the guide portion in a state where the cover according to the first embodiment is positioned at the first opened position and the holding portion has been operated.
- (13) FIG. 7B is a view illustrating the guide portion viewed from the other side of the guide portion in a state where the cover according to the first embodiment is positioned at the first opened position and the holding portion has been operated.
- (14) FIG. 8A is a view illustrating the guide portion viewed from one side of the guide portion in a state where the cover according to the first embodiment is positioned at the second opened position.
- (15) FIG. 8B is a view illustrating the guide portion viewed from the other side of the guide portion in a state where the cover according to the first embodiment is positioned at the second opened position.
- (16) FIG. 9A is a view illustrating the cover and the guide portion viewed from one side of the guide portion in a state where the cover according to the first embodiment is positioned at the closed position.
- (17) FIG. 9B is a view illustrating the cover and the guide portion viewed from the other side of the guide portion in a state where the cover according to the first embodiment is positioned at the closed position.
- (18) FIG. 10A is a view illustrating the cover and the guide portion viewed from one side of the guide portion in a state where the cover according to the first embodiment is positioned at the first opened position.

(19) FIG. 10B is a view illustrating the cover and the guide portion viewed from the other side of the guide portion in a state where the cover according to the first embodiment is positioned at the first opened position.

(20) FIG. 11A is a view illustrating the cover and the guide portion viewed from one side of the guide portion in a state where the cover according to the first embodiment is positioned at the second opened position.

(21) FIG. 11B is a view illustrating the cover and the guide portion viewed from the other side of the guide portion in a state where the cover according to the first embodiment is positioned at the second opened position.

(22) FIG. 12A is a rear view of a cover and a guide portion in a state where a cover according to a second embodiment is positioned at a first opened position.

(23) FIG. 12B is a left side view of the cover and the guide portion in a state where the cover according to the second embodiment is positioned at the first opened position.

(24) FIG. 12C is a front view of the cover and the guide portion in a state where the cover according to the second embodiment is positioned at the first opened position.

(25) FIG. 13A is a rear view of the cover and the guide portion in a state where the cover according to the second embodiment is positioned at the second opened position.

(26) FIG. 13B is a left side view of the cover and the guide portion in a state where the cover according to the second embodiment is positioned at the second opened position.

(27) FIG. 13C is a front view of the cover and the guide portion in a state where the cover according to the second embodiment is positioned at the second opened position.

(28) FIG. 14A is a front view of the cover in a state where the cover according to the second embodiment is positioned at the closed position.

(29) FIG. 14B is a front view of the guide portion in a state where the cover according to the second embodiment is positioned at the closed position.

(30) FIG. 15A is a front view of the cover in a state where the cover according to the second embodiment is positioned at the first opened position.

(31) FIG. 15B is a front view of the guide portion in a state where the cover according to the second embodiment is positioned at the first opened position.

(32) FIG. 16A is a front view of the cover illustrating the guide portion in a state where the cover according to the second embodiment is positioned at the first opened position and the holding portion has been operated.

(33) FIG. 16B is a front view of the guide portion illustrating the guide portion in a state where the cover according to the second embodiment is positioned at the first opened position and the holding portion has been operated.

(34) FIG. 17A is a front view of the cover in a state where the cover according to the second embodiment is positioned at the second opened position.

(35) FIG. 17B is a front view of the guide portion in a state where the cover according to the second embodiment is positioned at the second opened position.

(36) FIG. 18A is a front view of the cover illustrating an operation where the cover according to the second embodiment is returned from the second opened position to the first opened position.

(37) FIG. 18B is a front view of the guide portion illustrating an operation where the cover according to the second embodiment is returned from the second opened position to the first opened position.

(38) FIG. 19A is a front view of the cover illustrating an operation where the cover according to the second embodiment is returned from the first opened position to the closed position.

(39) FIG. 19B is a front view of the guide portion illustrating an operation where the cover according to the second embodiment is returned from the first opened position to the closed position.

(40) FIG. 20 is an exploded perspective view of a configuration related to an opening and closing

of a cover according to a third embodiment.

(41) FIG. 21 is an explanatory view of a link configuration of a switch according to the third embodiment.

(42) FIG. 22 is a front view of a state where the cover according to the third embodiment is positioned at a closed position.

(43) FIG. 23 is a front view of a state where the cover according to the third embodiment is positioned at a first opened position.

(44) FIG. 24 is a front view of an internal configuration in a state where the cover according to the third embodiment is positioned at the first opened position.

(45) FIG. 25 is a front view of the cover according to the third embodiment positioned at a second opened position.

(46) FIG. 26 is an explanatory view of a state where the cover according to the third embodiment is moved from the second opened position to the closed position.

(47) FIG. 27 is an explanatory view illustrating a configuration for preventing erroneous operation of the cover according to the third embodiment.

(48) FIG. 28 is an explanatory view in a state where the cover according to the third embodiment is moved from the first opened position to the second opened position.

(49) FIG. 29A is a front view of a cover according to a fourth embodiment in a state where a fixing member is in a first state.

(50) FIG. 29B is a front view of the cover according to a fourth embodiment in a state where the fixing member is in a second state.

(51) FIG. 30A is a front view of the cover according to a fourth embodiment in a state where the fixing member has been returned from the second state to the first state.

(52) FIG. 30B is a front view of the cover according to a fourth embodiment in a state where the fixing member has retreated from the first state.

(53) FIG. 31 is a front view of a cover according to a modified example of the fourth embodiment in which a fixing member is in a first state.

(54) FIG. 32A is a front view of a cover according to a fifth embodiment in which a fixing member has been returned from a second state to a first state.

(55) FIG. 32B is a front view of the cover according to the fifth embodiment in a state where a movable member of a fixing member has been rotated.

DESCRIPTION OF THE EMBODIMENTS

First Embodiment

(56) A first embodiment of the present invention will be described in detail below with reference to FIGS. 1 to 11. In the present embodiment, a tandem-type full-color printer is described as an example of an image forming apparatus 1. However, the present invention is not limited to a tandem-type image forming apparatus 1, and it may be other types of image forming apparatuses, and it may even be applied not only to full-color printers but also to monochrome, mono-color, and inkjet printers.

(57) Image Forming Apparatus

(58) At first, a general configuration of the image forming apparatus 1 according to a first embodiment will be described with reference to FIG. 1. FIG. 1 is a schematic drawing of a color image forming apparatus using an electrophotographic system. The image forming apparatus 1 is an intermediate transfer tandem-type image forming apparatus in which four color image forming units of yellow (Y), magenta (M), cyan (C), and black (Bk) are aligned above the intermediate transfer belt 2. A direction vertical to the sheet surface in FIG. 1 is a front-rear direction of the apparatus.

(59) The image forming apparatus 1 includes a casing 1a, and process cartridges Pa, Pb, Pc, and Pd stored in the casing 1a for forming toner images. The process cartridges Pa, Pb, Pc, and Pd are examples of an image forming portion and form an image on a sheet. The apparatus body includes

the casing **1a** and the process cartridges Pa, Pb, Pc, and Pd. The process cartridges Pa, Pb, Pc, and Pd use toner of different colors, which are yellow, magenta, cyan, and black, and they adopt similar configurations except for the different toner colors. Therefore, a process cartridge Pa of yellow toner is described below as a representative example.

(60) In FIG. **1**, a surface of a photosensitive member **6** that is driven to rotate in a clockwise direction is exposed and scanned by an exposing unit **7** driven based on an image information signal, and an electrostatic latent image is formed thereby. The electrostatic latent image is developed by the process cartridges Pa, Pb, Pc, and Pd as a toner image. Thereafter, predetermined pressurizing force and primary transfer bias are applied by a primary transfer apparatus **T1**, and a toner image is formed on the intermediate transfer belt **2**. A primary transfer bias roller **5** and a transfer cleaning apparatus **4** are in contact with the intermediate transfer belt **2**. Further, the intermediate transfer belt **2** is an endless belt that is stretched across a secondary transfer inner roller **9**, a tension roller **3**, and a secondary transfer upstream roller **8** that also function to transmit drive to the intermediate transfer belt **2**, and that is driven to be conveyed in a counterclockwise direction in the drawing.

(61) The primary transfer bias roller **5** forms a primary transfer nip by nipping the intermediate transfer belt **2** with the photosensitive member **6**. An image forming process is performed at a timing at which a toner image is superposed on a color toner image that has been primarily transferred upstream on the intermediate transfer belt **2** by sequentially passing through the primary transfer nip, along with the conveyance of the intermediate transfer belt **2**. Four color toner images are formed on the intermediate transfer belt **2** and conveyed to a secondary transfer unit **T2**.

(62) A sheet S, i.e., storage medium, that has been stacked and stored in a sheet cassette **11** is fed one by one by a feed roller **12**, and skewing of the sheet S is corrected by a registration roller **13** before the sheet S is transmitted at an appropriate timing to the secondary transfer unit **T2**. In the secondary transfer unit **T2**, a nip is formed by interposing the intermediate transfer belt **2** between the secondary transfer inner roller **9** and a secondary transfer outer roller **14**, and the four-color toner image formed on the intermediate transfer belt **2** is secondarily transferred to the sheet S. The sheet S after secondary transfer is conveyed to a fixing unit **15**, where the toner image on the sheet S is fixed, and the sheet S is discharged to an exterior of the apparatus by a sheet discharge roller pair **16** and sequentially stacked on a sheet discharge tray **17**.

(63) General Configuration of Opening/Closing Mechanism of Cover

(64) Next, an opening/closing mechanism of a cover **10** will be described. An outline of the opening/closing mechanism for setting an angle, an orientation, and a position of the cover **10** will be described. FIG. **2A** is an explanatory view of a closed position state, i.e., sheet conveyable state, or standby state, of the opening/closing mechanism according to the present embodiment, FIG. **3A** is an explanatory view of a first opened position, i.e., jammed sheet removable position, of the opening/closing mechanism, and FIG. **4A** is an explanatory view of a second opened position, i.e., unit replaceable position, of the opening/closing mechanism. FIGS. **2B**, **3B**, and **4B** respectively illustrate a position of a slide shaft **21** in a guide portion **30** corresponding to FIGS. **2A**, **3A**, and **4A**.

(65) Jamming of sheets S may occur within the conveyance path at a right side of the image forming apparatus **1**, such that the cover **10** that may be used to open and close this part of the apparatus is disposed on the casing **1a**. The cover rotation shaft **18** is provided on the cover **10**, and by having the image forming apparatus **1** engage with the cover rotation shaft **18**, the cover **10** is supported rotatably about a horizontal axis in a front-rear direction. That is, according to the present embodiment, the cover **10** constitutes a right side wall portion of the casing **1a** in the closed position, and it is disposed rotatably about the cover rotation shaft **18** in the horizontal direction at a lower part of the cover **10**. The cover **10** is designed to have its angle, position, and orientation determined by a support arm **20**, and the cover **10** moves arbitrarily among three positions, which are a closed position, a first opened position, and a second opened position. The closed position is a

position at which the cover **10** is stored in the image forming apparatus **1**, which is referred to as a sheet conveyable state or a standby state. In the closed position, the cover **10** closes the casing **1a** and forms a sheet conveyance path. The first opened position is a position at which the operator may remove jammed sheets, which is referred to as a jammed sheet removable position. The second opened position is a position at which the operator may replace units, which is referred to as a unit replaceable position. That is, the cover **10** is disposed in an openable and closable manner by rotating about the casing **1a**, and it is movable between a first opened position that is inclined by a first angle with respect to a closed position where the casing **1a** is closed to expose the sheet conveyance path, and a second opened position that is inclined by a second angle greater than the first angle with respect to the closed position to expose the sheet conveyance path.

(66) The support arm **20** has the slide shaft **21** disposed on a first end portion so as to connect the support arm **20** and the cover **10** and is rotatably supported at a second end portion by a rotation shaft **22** to the casing **1a** of the image forming apparatus **1**. The cover **10** includes the guide portion **30** that guides the support arm **20**. The guide portion **30** includes two paths, which are a first path **31**, i.e., first area, and a second path **32**, i.e., second area. That is, the guide portion **30** is an example of a guide member disposed on the cover **10**, and includes the first path **31** which is an example of a first guide and the second path **32** which is an example of a second guide. The first path **31** guides the support arm **20** such that the cover **10** is positioned at the closed position and at the first opened position. The second path **32** is branched from the first path **31**, and guides the support arm **20** such that the cover **10** is positioned at the second opened position.

(67) An angle, i.e., position or orientation, of the support arm **20** is determined by the position of the slide shaft **21** on the guide portion **30**, and simultaneously, an angle of the cover **10** is also determined. That is, the support arm **20** is an example of a support member that is disposed swingably on the casing **1a**, that includes the slide shaft **21** serving as a guided portion guided by the guide portion **30**, and that can support the cover **10** in an inclined state on the casing **1a**. Further, the support arm is an example of a connecting member that is interlocked with the opening and closing operation of the cover **10**, and that is connected to the casing **1a** and the cover **10**.

(68) Closed Position, i.e., Sheet Conveyable State or Standby State

(69) As illustrated in FIGS. 2A and 2B, when the slide shaft **21** of the support arm **20** is positioned on an end portion **31a** of the first path **31**, the support arm **20** is approximately parallel to the cover **10** in an approximately vertical direction, and the cover **10** is in a closed position stored in the image forming apparatus **1**. In this state, by providing a lock mechanism (not shown) to an appropriate position by which the cover **10** is locked to the image forming apparatus **1**, the slide shaft **21** will not receive any load from the cover **10**. In the closed position, which is the position taken during most of the period of use of the image forming apparatus **1**, the slide shaft **21** will not receive load from the cover **10**, such that it is not vulnerable to deterioration over time, such as creep deformation.

(70) First Opened Position, i.e., Jammed Sheet Removable Position

(71) As illustrated in FIGS. 3A and 3B, when the slide shaft **21** of the support arm **20** is positioned at an end portion **31b** opposite to the closed position of the first path **31**, the support arm **20** is in a balanced state by the rotation shaft **22** and the slide shaft **21** positioned at the end portion **31b**. The cover **10** is positioned at the first opened position by the cover rotation shaft **18** and the slide shaft **21** positioned at the end portion **31b** being balanced. The first opened position is required to provide an open space through which the operator can remove jammed sheets, and a first angle which is an opening angle thereof is preferably set, for example, to an angle of 45 degrees or greater and smaller than the angle of a second opened position. In the present embodiment, the first angle is set to 45 degrees. The first angle is not limited to 45 degrees or greater, and it can be set within the range of 30 to 60 degrees, for example.

(72) Second Opened Position, i.e., Unit Replaceable Position

(73) As illustrated in FIGS. 4A and 4B, when the slide shaft **21** of the support arm **20** is positioned

at an end portion **32a** of the second path **32**, the support arm **20** is in a balanced state by the rotation shaft **22** and the slide shaft **21** positioned at the end portion **32a**. The cover **10** is positioned at the second opened position by the balance between the cover rotation shaft **18** and the slide shaft **21** positioned at the end portion **32a**. The second opened position is required to provide an open space through which the operator can replace units of the image forming apparatus **1** and parts of the cover **10**, and a second angle which is an opening angle thereof is preferably set to an angle of 70 degrees or greater. In the present embodiment, the second angle is set to 75 degrees. The second angle is not limited to 70 degrees or greater, and it may be set within the range of 60 to 90 degrees, and may even be greater than 90 degrees.

(74) Path Switching Mechanism

(75) Next, a path switching mechanism **50** that switches the first path **31** and the second path **32** of the guide portion **30** will be described. According to the present embodiment, path switching mechanisms **50** are provided on both ends in the front and rear directions of the cover **10**. However, since the configurations thereof are similar, only one of the path switching mechanisms **50** will be described. The path switching mechanism **50** may also be provided on only a front end portion of the cover **10**.

(76) FIGS. **5A** to **8B** are explanatory views showing a first path opened state of the path switching mechanism **50** according to the present embodiment. The second path **32** is formed to branch from a side surface of the first path **31**. A path switching member **25** for switching paths between the first path **31** and the second path **32** is arranged at the branching point. The path switching member **25** is arranged rotatably about a rotation shaft **26** disposed on the cover **10**, and it is engaged rotatably by an elastic engagement pawl **10a** disposed on the cover **10**. That is, the path switching mechanism **50** is an example of a rotating and switching portion, and the cover **10** may be switched between a first state in which the cover **10** is rotatable between the closed position and the first angle and a second state in which the cover **10** is rotatable at least between the first angle and the second angle.

(77) The path switching mechanism **50** includes the guide portion **30**, the support arm **20**, and the path switching member **25**. The guide portion **30** includes the first path **31** and the second path **32**. The first path **31** enables the cover **10** to rotate between the closed position and the first angle by guiding the slide shaft **21** therein. The second path **32** enables the cover **10** to rotate from an angle exceeding the first angle to the second angle by guiding the slide shaft **21** therein.

(78) The path switching member **25** is an example of a regulation portion capable of regulating movement of the slide shaft **21**, and includes a first path blocking portion **25a** that blocks the first path **31** and a second path blocking portion **25b** that blocks the second path **32**. In a state where the first path blocking portion **25a** protrudes into the path of the guide portion **30**, the first path **31** is blocked and the second path **32** is opened, such that the slide shaft **21** is enabled to move in the second path **32**. Meanwhile, in a state where the second path blocking portion **25b** protrudes into the path, the first path **31** is opened, such that the slide shaft **21** is enabled to move in the first path **31**. That is, according to the path switching member **25**, in a state where the slide shaft **21** is positioned on the first path **31**, by having the second path blocking portion **25b** protrude into the path and restrict entry to the second path **32**, the path switching mechanism **50** is set to a first state (refer to FIGS. **5A** to **6B**). Moreover, according to the path switching member **25**, in a state where the slide shaft **21** is positioned at the end portion **31b** of the first path **31**, by having the first path blocking portion **25a** protrude into the path to communicate the first path **31** and the second path **32**, the path switching mechanism **50** is set to a second state (refer to FIGS. **7A** to **8B**). In a state where the path switching mechanism **50** is in the second state, the first path blocking portion **25a** serves as an example of a contact portion that intersects a trajectory of the slide shaft **21** in the first path **31** of the guide portion **30**. As described, the support arm **20** may be switched between a first state in which the cover **10** is rotatable between the closed position and the first opened position and a second state in which the cover **10** is rotatable at least between the first opened state and the second opened state. A holding portion **25c** described later is used to switch the support arm **20**

between the first state and the second state.

(79) In order to enable the path switching member **25** to switch paths without fail, a mechanism to hold the position stably at each of a first path opening position where the first path **31** is opened and a second path opening position where the second path **32** is opened is required. According to the present embodiment, at first, in order to stabilize the position at the first path opening position, an engagement pawl **27** is provided on the path switching member **25**, which is engaged with an engagement portion **28** of the cover **10** in a state where the path switching member **25** is positioned at the first path opening position. The engagement portion **28** is formed to protrude from an inner side of the cover **10**. That is, the engagement portion **28** is an example of a retaining portion that retains the path switching member **25** such that the path switching mechanism **50** is maintained at the first state. Further, the path switching member **25** is released from retention by the engagement portion **28** through the operation of a holding portion **25c**. A pawl member that deforms elastically is adopted as the engagement pawl **27** of the present embodiment considering costs, but the present technique is not limited thereto, and a configuration may be adopted in which a member that is different from the path switching member **25** and that is moved in sliding motion by an elastic spring member may be provided, as long as it has repeatability. Simultaneously, in order to stabilize the position at the second path opening position, the engagement pawl **27** is engaged with an engagement portion **29** of the cover **10** in a state where the path switching member **25** is positioned at the second path opening position (refer to FIG. 7B).

(80) The holding portion **25c** that is used by the operator when performing a path switching operation is provided on the path switching member **25**. That is, the holding portion **25c** is one example of an operating portion for switching the path switching mechanism **50** between the first state and the second state. In a state where the cover **10** is positioned at the first opened position, the holding portion **25c** is operated by the user to be moved from a regulating position in which the support arm **20** is regulated from moving from the first path **31** to the second path **32** to an allowing position in which the support arm **20** is allowed to move from the first path **31** to the second path **32**. In the present embodiment, the holding portion **25c** may be moved mutually between the regulating position and the allowing position. The operator operates the holding portion **25c** to open the second path **32** when switching the cover **10** from the first opened position to the second opened position, and guides the path switching member **25** to the second path opening position. The switching operation is completed by engaging the engagement pawl **27** to the engagement portion **28** of the cover **10**. The holding portion **25c** is exposed toward an inner side from an inner wall of the cover **10**, and it is provided at a part other than the part being an external wall of the casing **1a** at the cover **10** in the closed position. Therefore, it becomes possible to prevent the user from operating the holding portion **25c** unintendedly when the cover **10** is closed.

(81) Meanwhile, in a state where the first path **31** is opened, the operator rotates the cover **10** to the closed position without operating the holding portion **25c**. Thereby, switching is completed by the slide shaft **21** coming into contact with the first path blocking portion **25a**, guiding the path switching member **25** by pushing the same to the first path opening position, and engaging the engagement pawl **27** to the engagement portion **28** of the cover **10**. By adopting the above-mentioned configuration, a switching mechanism for switching the paths of the slide shaft **21** can be realized. That is, in a state where the holding portion **25c** is at the regulating position, the support arm **20** and the first path **31** are engaged, and which the cover **10** is retained at the first opened position. Further, in a state where the holding portion **25c** is in the allowing position, the support arm **20** is moved from the first path **31** to the second path **32**, and the cover **10** is retained at the second opened position.

(82) Opening/Closing Operation of Cover

(83) An operation by which the operator switches the opening/closing angle of the cover **10** and a movement of the mechanism will be described. FIGS. 9A and 9B are an explanatory view of the opening/closing mechanism in the closed position state and the path switching mechanism **50**

according to the present embodiment. FIGS. **10A** and **10B** are explanatory views of the opening/closing mechanism in the first opened position and the path switching mechanism **50**. FIGS. **11A** and **11B** are explanatory views of the opening/closing mechanism in the second opened position and the path switching mechanism **50**.

(84) From Closed Position to First Opened Position

(85) In order to open the cover **10** from the closed position, i.e., stored position, to the first opened position, i.e., jammed sheet removable position, if a lock mechanism is provided to lock the cover **10** to the image forming apparatus **1**, the lock may be unlocked. When the opening operation is started, the slide shaft **21** of the support arm **20** slides along the first path **31** from an end portion **31a** toward the end portion **31b**. In this state, the path switching member **25** is at the first path opening position and the second path **32** is blocked, such that the slide shaft **21** is guided to the slide shaft **21** and reaches the end portion **31b**, and the support arm **20** and the cover **10** are stopped in the balanced state. Thus, the opening operation to the first opened position, i.e., jammed sheet removable position, is completed.

(86) From First Opened Position to Closed Position

(87) When closing the cover **10** from the first opened position, i.e., jammed sheet removable position, to the closed position, i.e., stored position, the operator pushes the cover **10** toward the casing **1a** of the image forming apparatus **1** using his/her hand. Thereby, the slide shaft **21** of the support arm **20** slides on the first path **31** from the end portion **31b** toward the end portion **31a**, and the lock mechanism is locked to the image forming apparatus **1**, by which the movement is completed.

(88) From First Opened Position to Second Opened Position

(89) When opening the cover **10** from the first opened position, i.e., jammed sheet removable position, to the second opened position, i.e., unit replaceable position, at first, the path switching member **25** is moved to the second path opening position. The operator operates the holding portion **25c** to move the same to the second path opening position, and the engagement pawl **27** is engaged from the engagement portion **28** to the engagement portion **29** of the cover **10**. Thereby, the first path **31** is blocked, and instead, the second path **32** is opened. In this state, the cover **10** and the support arm **20** are still balanced and are still maintained at the first opened position. In this state, when the operator lifts the cover **10** slightly by which the slide shaft **21** is separated from the end portion **31b**, the balanced state is cancelled and the slide shaft **21** is guided to the second path **32**.

(90) In this state, the cover **10** is not opened to the second opened position, i.e., unit replaceable position, immediately after switching the path switching member **25**, and the balanced state is maintained. In other words, the path switching member **25** merely blocks the second path **32**, and the load of the cover **10** is received by the end portion **31b** provided on the first path **31**. That is, the end portion **31b** is an example of a receiving portion that receives the load of the cover **10** acting on the slide shaft **21** when the path switching mechanism **50** is in the first state and the cover **10** is positioned at the first opened position. Thereby, the only operating force for operating the path switching member **25** is the load that occurs when engaging the engagement pawl **27** from the engagement portion **29** to the engagement portion **28** of the cover **10**. Thereby, the operating force exerted by the operator when performing operation can be reduced, and the life of the product can be extended. The slide shaft **21** guided to the second path **32** reaches the end portion **32a**, and the support arm **20** and the cover **10** are stopped at the balanced state. Thereby, the opening operation to the second opened position, i.e., unit replaceable position, is completed.

(91) From Second Opened Position to Closed Position

(92) When closing the cover **10** from the second opened position, i.e., unit replaceable position, to the closed position, i.e., stored position, the operator pushes the cover **10** toward the casing **1a** of the image forming apparatus **1** using his/her hand. Thereby, the slide shaft **21** of the support arm **20** slides on the second path **32** and the first path **31** from the end portion **32a** toward the end portion

31a, and the lock mechanism is locked to the image forming apparatus **1**, by which the operation is completed. In this state, the first path **31** is blocked by the path switching member **25**, but the slide shaft **21** is in contact with the first path blocking portion **25a** and pushes the path switching member **25** to guide the same toward the first path opening position. Then, switching is completed by engaging the engagement pawl **27** from the engagement portion **29** to the engagement portion **28** of the cover **10**. That is, in a state where the slide shaft **21** is positioned at the second path **32**, the cover **10** rotates toward the closed position, by which the slide shaft **21** abuts against and pushes the first path blocking portion **25a**, such that the path switching mechanism **50** is switched from the second state to the first state.

(93) Thereby, the operation from the closed position, i.e., stored position, to the first opened position, i.e., jammed sheet removable position, may be restarted. Thereby, when closing the cover **10** from the second opened position to the closed position, a configuration is adopted in which the path switching member **25** is returned to the first path opening position in a manner interlocked with the operation of the cover **10** without the operator operating the holding portion **25c**. That is, the path switching mechanism **50** is switched from the first state to the second state by the operation of the holding portion **25c**, while the path switching mechanism **50** is switched from the second state to the first state by rotating the cover **10** from the second angle to the closed position. Thereby, the possibility of the operator performing an erroneous operation, such as failure to operate the mechanism, may be reduced.

(94) As described, according to the image forming apparatus **1** of the present embodiment, the path switching mechanism **50** includes the holding portion **25c** that switches between the first state and the second state. Therefore, the cover **10** may be opened greatly from the first angle to the second angle by the operation of the holding portion **25c**, such that the operability for opening the cover **10** in two stages can be simplified. That is, since the path switching operation of opening the cover **10** from the first stage to the second stage can only be realized by switching the holding portion **25c**, a simultaneous operation of receiving the load of the cover **10** while operating the support arm to switch paths becomes unnecessary. Thereby, the operation for opening the cover to the second stage is simplified, such that the switching operation may be performed by anyone, instead of being limited to a dedicated operator.

(95) According to the embodiment mentioned above, a case has been described in which the cover **10** is a cover disposed on a right side of the casing **1a**, but the present technique is not limited thereto. For example, the cover may be disposed on the left side of the casing **1a**, or the cover may be a front cover. According further to the present embodiment, a case has been described in which the cover rotation shaft **18** supporting the cover **10** is arranged in a horizontal direction, but the present technique is not limited thereto, and for example, the cover rotation shaft may be arranged in the vertical direction.

(96) According to the embodiment described above, a case has been described in which the cover **10** is opened in two stages, but the present technique is not limited thereto, and a configuration can be adopted in which the cover **10** is opened in three or more stages. In that case, paths for the guide portion per stage and path switching members are provided.

(97) According to the embodiment described above, a case has been described in which the cover **10** is a cover of the image forming apparatus **1** equipped with an image forming unit, but the present technique is not limited thereto, and the image forming apparatus may include a postprocessing apparatus such as a finisher, and the cover may be a casing of the postprocessing apparatus.

Second Embodiment

(98) Next, a second embodiment of the present invention will be described in detail with reference to FIGS. **12A** to **19**. The present embodiment differs from the first embodiment in that path switching mechanisms **150** disposed at each end portion in the front-rear direction of the cover **10** are interlocked. Other configurations are similar to the first embodiment, such that the same

reference numbers are assigned to corresponding components, and detailed descriptions thereof are omitted.

(99) According to the present embodiment, as illustrated in FIGS. **12** and **13**, path switching mechanisms **150A** and **150B** are respectively arranged on end portions in the front-rear direction of the cover **10**. A configuration of an interlocking portion **33** that interlocks the path switching mechanisms **150A** and **150B** will be described later. The path switching mechanisms **150A** and **150B** adopt a symmetrical shape, and configurations thereof are similar, such that only one of the path switching mechanisms **150** will be described below with reference to FIGS. **14A** to **19B**.

(100) General Configuration of Opening/Closing Mechanism of Cover

(101) FIG. **14A** is an explanatory view of a closed position state, i.e., sheet conveyable state or standby state, of an opening/closing mechanism according to the present embodiment, FIG. **15A** is an explanatory view of a first opened position, i.e., jammed sheet removable position, of the opening/closing mechanism, and FIG. **17A** is an explanatory view of a second opened position, i.e., unit replaceable position, of the opening/closing mechanism. FIGS. **14B**, **15B**, and **17B** respectively illustrate the position of a slide shaft **21** in a guide portion **30** corresponding to FIGS. **14A**, **15A**, and **17A**.

(102) Closed Position, i.e., Sheet Conveyable State or Standby State

(103) As illustrated in FIGS. **14A** and **14B**, in a state where the slide shaft **21** of a support arm **20** is positioned at an end portion **131a** of a first path **131**, the support arm **20** is approximately parallel to approximately a vertical direction of the cover **10**, and the cover **10** is at a closed position stored in the image forming apparatus **1**.

(104) First Opened Position, i.e., Jammed Sheet Removable Position

(105) As illustrated in FIGS. **15A** and **15B**, when the slide shaft **21** of the support arm **20** is positioned at an end portion **131b** opposite to the closed position of the first path **131**, the support arm **20** is in a balanced state by the rotation shaft **22** and the slide shaft **21** positioned at the end portion **131b**. The cover **10** is positioned at the first opened position by the cover rotation shaft **18** and the slide shaft **21** positioned at the end portion **131b** being balanced.

(106) Second Opened Position, i.e., Unit Replaceable Position

(107) As illustrated in FIGS. **17A** and **17B**, when the slide shaft **21** of the support arm **20** is positioned at an end portion **132a** of a second path **132**, the support arm **20** is in a balanced state by the rotation shaft **22** and the slide shaft **21** positioned at the end portion **132a**. The cover **10** is positioned at the second opened position by the cover rotation shaft **18** and the slide shaft **21** positioned at the end portion **132a** being balanced.

(108) Operation of Path Switching Mechanism

(109) An operation of a guide portion **130** when the cover **10** is moved from the first opened position to the second opened position will be described. In order to move the cover **10** from the first opened position to the second opened position, it is necessary to move the slide shaft **21** from the first path **131** to the second path **132**. From the closed position illustrated in FIGS. **14A** and **14B**, the operation of moving the slide shaft **21** from the first path **131** to the second path **132** and moving the cover **10** from the first opened position to the second opened position will be described. In a state where the slide shaft **21** is positioned at the end portion **131b**, the operator moves a path switching member **125** from a position **125a** to a position **125b** illustrated in FIG. **16**, and the second path **132** is opened. Thereafter, as illustrated in FIG. **17A**, by closing the cover **10** slightly, the slide shaft **21** falls into the second path **132**, and in that state, by opening the cover **10** again, the cover **10** may be opened to the second opened position.

(110) Operation of Path Switching Member During Opening Operation of Cover

(111) FIG. **12** is a three-side view illustrating the interlock of the path switching mechanisms **150A** and **150B** provided on both ends in a rotation center shaft direction of the cover **10**. In image forming apparatuses **1**, the size of the cover **10** tends to increase along with the increase in corresponding sheet sizes, the enhancement of speed and the increase of functions leading to

improved productivity. In the present embodiment, in order to regulate the opening angle of the cover **10** that may be large and heavy, the support arm **20** is provided on both end portions in the rotational axis direction of the cover **10**. Along therewith, guide portions **130A** and **130B** and path switching members **125A** and **125B** are provided respectively to correspond to the two support arms **20**. If slide shafts **21A** and **21B** of the two support arms **20** are positioned at different paths, deformation and damage may occur due to the twisting or distortion of the cover, or damage of the support arm **20** may be caused by the concentration of load on only one of the support arms **20**. In some cases, the cover itself may fall from the casing **1a** and the cover may be damaged. In order to prevent these drawbacks, it is desirable that the two path switching members **125A** and **125B** are interlocked when switching positions.

(112) The interlocking of operations of the path switching members **125A** and **125B** using the interlocking portion **33** will be described with reference to FIGS. **12A** to **12C**. The present embodiment adopts a configuration in which, by mechanically connecting the path switching members **125A** and **125B** using the interlocking portion **33**, when the path switching member **125A** is moved, the path switching member **125B** is also moved infallibly in an interlocked manner. The interlocking portion **33** includes a first link **34** and a second link **35** that are mutually connected rotatably. The first link **34** includes a first connecting portion that is interlocked with the path switching member **125A**, a second connecting portion that is connected rotatably to the second link **35**, and a first supporting portion **34a** that is supported rotatably to the cover **10** between the first connecting portion and the second connecting portion. The second link **35** includes a third connecting portion that is interlocked with the path switching member **125B**, a fourth connecting portion that is connected rotatably to the second connecting portion of the first link **34**, and a second supporting portion **35a** that is supported rotatably to the cover **10** between the third connecting portion and the fourth connecting portion.

(113) As illustrated in FIG. **12**, when the path switching member **125A** is positioned at the position **125a** (refer to FIG. **16**), the path switching member **125B** is also positioned at the position **125a** through the interlocking portion **33**. As illustrated in FIG. **13**, when the path switching member **125A** is positioned at the position **125b** (refer to FIG. **16**), the path switching member **125B** is also positioned at the position **125b** through the interlocking portion **33**. As described, the position of the path switching member **125B** can be switched infallibly by being interlocked with the switching of position of the path switching member **125A**.

(114) Operation of Path Switching Member for Cover Closing Operation, i.e., from Second Opened Position to First Opened Position or Closed Position

(115) The closing operation of the cover **10** from the second opened position to the first opened position and to the closed position state will be described with reference to FIGS. **18** and **19**. FIG. **18B** illustrates a state in which the cover **10** is closed from the second opened position and the slide shaft **21** has returned from the second path **132** to the end portion **131b** of the first path **131**. Further, FIG. **19** illustrates a state in which the closing operation is further advanced. According to the angle of the cover **10**, the path switching member **25** may be moved from the position **125b** to the position **125a** while the slide shaft **21** moves within the first path **131** toward the end portion **131a** of the first path **131**. Therefore, when an operation to close the cover **10** from the second opened position to the closed position is performed, the operator may return the path switching member **25** to the position **125a** simply by closing the cover **10**.

(116) As described, in the image forming apparatus **1** of the present embodiment, according to the opening/closing angle switching operation of the cover **10**, the position of the path switching member **125B** may be moved in an interlocked manner with the movement of position of the path switching member **125A**, such that the operation can be simplified. Therefore, not only dedicated operators but also general users can perform the switching operation, and replacement of periodically replaced units can be performed by the users.

Third Embodiment

(117) Next, a third embodiment of the present invention will be described in detail with reference to FIGS. 20 to 28. The configuration of the present embodiment differs from the first embodiment in that a switch 225 serving as an operating portion may be locked to prevent erroneous operation. Other configurations are similar to the first embodiment, such that the same reference numbers are assigned to corresponding components, and detailed descriptions thereof are omitted.

(118) Configuration Regarding Opening and Closing of Cover

(119) FIG. 20 illustrates a configuration regarding the opening and closing of the cover 10. The cover 10 includes a guide portion 230 having an opened shape, the switch 225, the cover rotation shaft 18, and the support arm 20. The support arm 20 includes the slide shaft 21 and the rotation shaft 22. The guide portion 230 and the slide shaft 21 are designed to be engaged, the cover rotation shaft 18 and the image forming apparatus 1 are designed to be engaged, and the rotation shaft 22 and the image forming apparatus 1 are designed to be engaged. The switch 225 is slidably attached to a switch guide 10c, which are connected through the lock portion 224 and a link 226 as illustrated in FIG. 21, wherein the link 226 is engaged rotatably with each of the switch 225 and the lock portion 224. The lock portion 224 is rotatably engaged to the cover 10 through a lock portion rotation shaft 10d. The cover 10 rotates about the cover rotation shaft 18, and the orientation of the cover 10 is determined by the slide shaft 21 and the guide portion 230.

(120) Transition from Closed Position to First Position

(121) FIG. 22 illustrates a detail of a closed position of the cover 10. When the slide shaft 21 of the support arm 20 is positioned at an end portion 231a of a first path 231, the support arm 20 is approximately parallel to the cover 10, and the cover 10 will be in a closed position stored in the image forming apparatus 1. In this state, by providing a lock mechanism by which the cover 10 is locked to the image forming apparatus 1 at an arbitrary position, the slide shaft 21 will not receive any load from the cover 10. When the cover 10 in the closed position state is opened, the slide shaft 21 is guided along the first path 231 of the guide portion 230 and moves from the end portion 231a to an end portion 231b, by which the cover 10 is transited to the first opened position illustrated in FIG. 23. When the slide shaft 21 of the support arm 20 is positioned at the end portion 231b of the first path 231, the support arm 20 is in a balanced state by the rotation shaft 22 and the slide shaft 21 positioned at the end portion 231b. The cover 10 is in a first opened position state by the cover rotation shaft 18 and the slide shaft 21 positioned at the end portion 231b being balanced.

(122) Transition from First Opened Position to Second Opened Position

(123) As illustrated in FIG. 23, in a state where the cover 10 is in a first opened position, the slide shaft 21 is regulated from being guided toward a second path 232 by the lock portion 224 at a branch portion 230a of the guide portion 230. When the switch 225 is operated toward an arrow D1 direction from this state, as illustrated in FIG. 24, a retained state is realized by a switch pawl portion 225a being engaged to a projected portion 10b of the cover 10. Further, by operating the switch 225, a moment of rotation in a D3 direction occurs about the lock portion rotation shaft 10d to the lock portion 224 through the link 226, and the lock portion 224 moves to a position where the slide shaft 21 is not regulated. A first spring 227 is a compression coil spring which is disposed to constantly urge the switch 225 toward the direction D2 to ensure the regulated state of the lock portion 224, and it is desirably provided. Alternatively, as a unit other than the spring, a configuration may be adopted to elastically engage the switch 225 and the cover 10.

(124) A second spring 228 is provided with the aim to aid the operation of the switch 225, by which a moment of rotation in a direction D3 occurs to the lock portion 224, which applies a force in a D1 direction to the switch 225 through the link 226, and it is preferably provided to reduce the force necessary for operation. In this state, the cover 10 and the support arm 20 are in a balanced state and stopped at a first opened position, as illustrated in FIG. 28. By lifting the cover 10 slightly by which the slide shaft 21 is separated from the end portion 231b, the state of balance is released and the slide shaft 21 is guided toward the second path 232.

(125) Now, the cover 10 will not be guided to the second path 232 immediately after switching the

lock portion **224**, and the balanced state is maintained. That is, the lock portion **224** merely blocks the second path **232**, and the load of the cover **10** is received by the end portion **231b** provided on the first path **231**. Thereby, the only operating force when operating the lock portion **224** is the load that occurs when engaging the switch pawl portion **225a** to the projected portion **10b** of the cover **10**. By adopting such configuration, the operating force that the user exerts during operation may be reduced.

(126) As illustrated in FIG. **25**, the slide shaft **21** guided to the second path **232** reaches an end portion **232a**, and the support arm **20** and the cover **10** are stopped in the balanced state. Thus, transition to the second opened position is completed. When the slide shaft **21** of the support arm **20** is positioned at the end portion **232a** of a second guide portion, the support arm **20** is in a balanced state by the rotation shaft **22** and the slide shaft **21** positioned at the end portion **232a**. The cover **10** is in a second opened state by the cover rotation shaft **18** and the slide shaft **21** positioned at the end portion **232a** being balanced.

(127) Transition from First and Second Opened Position to Closed Position

(128) When the user closes the cover **10** from the first opened position illustrated in FIG. **23**, the slide shaft **21** is guided by the guide portion **230** to move to the end portion **231a** and to the closed position. Further, if the user similarly closes the cover **10** from the second opened position illustrated in FIG. **25**, the slide shaft **21** is guided by the guide portion **230** to be passed through the branch portion **230a** and abut against the lock portion **224**. When the cover **10** is closed further from this state, as illustrated in FIG. **26**, a moment of rotation in a D4 direction occurs to the lock portion **224**, along with which the switch **225** and the link **226** are returned to their initial positions in an interlocked manner, and the slide shaft **21** is guided by the guide portion **230** to the end portion **231a** and transited to the closed position. As described, since the cover **10** may be transited to the closed position by simply closing the cover **10**, the action of the user operating the switch **225** before closing the door can be omitted.

(129) Configuration of Measures Against Erroneous Operation in Second Opened Position

(130) In the present embodiment, in a state where the cover **10** is in the second opened position illustrated in FIG. **25**, if the switch **225** is operated in the D2 direction toward the returning direction, a moment of rotation in the D3 direction as illustrated in FIG. **26** occurs, such that the lock portion **224** will not return to the initial position. If it is assumed that the switch **225** has returned to the initial position, the state of FIG. **27** is realized. That is, the lock portion **224** will be positioned in midway of the guide portion **230** forming a wall. In that case, if the user closes the cover **10** without performing an operation to return the switch, the slide shaft **21** and the lock portion **224** will collide against each other and may be damaged, so that the above-mentioned configuration is adopted to prevent such a state.

(131) As described, according to the image forming apparatus **1** of the present embodiment, when switching the cover **10** from the first opened position to the second opened position, the switching is enabled through a switching operation. If the user operates the switch at an unintended timing, the switch **225** will not be returned, and the cover **10** is moved to the closed position only through the operation of closing the cover **10**, such that that intuitive and safe operation is enabled.

Fourth Embodiment

(132) Next, a fourth embodiment of the present invention will be described in detail with reference to FIGS. **29A** to **31**. The configuration of the present embodiment differs from the first embodiment in that a function of retreating a fixing member **40** from a first state is provided. Other configurations are similar to the first embodiment, such that the same reference numbers are assigned to corresponding components, and detailed descriptions thereof are omitted.

(133) Configuration Related to Opening and Closing of Cover

(134) The support arm **20** includes the slide shaft **21** connecting the support arm **20** and the cover **10**, and the rotation shaft **22** connecting the support arm **20** and the image forming apparatus **1**. The cover **10** is provided with a guide portion **330** that guides the support arm **20**, and the angle, i.e.,

position and orientation, of the support arm **20** is determined by the position of the slide shaft **21** on the guide portion **330**, and simultaneously, the opening angle of the cover **10** is determined.

(135) FIG. **29A** is an enlarged view illustrating from a front side of the apparatus the cover **10** in a state opened to the first angle. As illustrated in FIG. **29A**, the fixing member **40** is provided to retain the cover **10** at the first angle. The fixing member **40** is provided on the cover **10** and the fixing member **40** contacts the slide shaft **21** guided by the guide portion **330** by a support surface **43**, and retains the slide shaft **21** at the position illustrated in FIG. **29A**. The fixing member **40** is supported on the cover **10** in a manner rotatable from the state illustrated in FIG. **29A**, i.e., first state, about a fixing member rotation shaft **41**, and the fixing member **40** is retained by a rotation stop portion not shown so as not to rotate in a counterclockwise direction. Accordingly, the support arm **20** is in a state balanced by the rotation shaft **22** and the slide shaft **21** in contact with the support surface **43**, and the cover **10** is also retained at the first angle by the cover rotation shaft **18** and the slide shaft **21** in contact with the support surface **43** being balanced.

(136) FIG. **29B** is an enlarged view illustrating from a front side of the apparatus a state in which the fixing member **40** is moved and the cover **10** is transited from the first angle to the second angle. If the slide shaft **21** of the support arm **20** is moved to an end portion of the guide portion **330** illustrated in FIG. **29B**, the support arm **20** is in a balanced state by the rotation shaft **22** and the slide shaft **21**. The cover **10** is retained at the second angle by the cover rotation shaft **18** and the slide shaft **21** positioned at the end portion of the guide portion **330** being balanced. The fixing member **40** has the state of FIG. **29B** set as the second state.

(137) In order to open the cover **10** from the first angle to the second angle, that is, in order to move the slide shaft **21**, the rotation stop portion, i.e., rotation lock, not shown of the fixing member **40** should be moved by the operator. In this state, the rotation stop portion may be movably supported on the cover **10** or may be removably disposed on the cover **10**. As described above, by moving the rotation stop portion, the fixing member **40** will be in a state rotatable in the counterclockwise direction, and along with the rotation of the fixing member **40**, the slide shaft **21** is moved within the guide portion **330** to a position where the cover **10** is retained at the second angle. In the present embodiment, after the slide shaft **21** moves beyond the fixing member **40** toward the end portion of the guide portion **330** illustrated in FIG. **29B**, the fixing member **40** is designed be returned automatically to the first state illustrated in FIG. **29A**.

(138) Closing Operation from Second Position

(139) FIG. **30A** is an enlarged view illustrating from a front side of the apparatus a movement of the slide shaft **21** when the cover **10** is closed from the second angle to the first angle, and FIG. **30B** is an enlarged view illustrating from the front side of the apparatus a movement of the slide shaft **21** and the fixing member **40** when the cover **10** is closed from the second angle to the first angle. Actually, when the slide shaft **21** is moved, the movement of the slide shaft **21** is interlocked with the movement in which the cover **10** is closed from the second angle to the first angle, such that the angle is further closed toward the body of the image forming apparatus **1** than the opening angle of the cover **10** illustrated in FIG. **30A**. However, the movement of the slide shaft **21** and the fixing member **40** is relevant to the present embodiment, such that further movement of the cover **10** is omitted.

(140) In the present embodiment, after the slide shaft **21** passes the fixing member **40** toward the end portion of the guide portion **330** illustrated in FIG. **29B**, the fixing member **40** is designed to be automatically returned to the state illustrated in FIG. **29A**. In this case, in a general conventional configuration, as illustrated in FIG. **30A**, the slide shaft **21** comes into contact with a non-support surface **44** of the fixing member **40** and the fixing member **40** will not move, such that the slide shaft **21** is blocked in this position. If the cover **10** is forcibly closed from this state, the fixing member **40**, the support arm **20**, or the cover **10** may be deformed or damaged.

(141) Therefore, according to the present embodiment, as illustrated in FIG. **30B**, the fixing member **40** is supported rotatably further in the clockwise direction from the state of FIG. **30A**.

Thereby, when the slide shaft **21** moves from the state of FIG. **30A** toward the closing direction of the cover **10**, the non-support surface **44** is pushed by the slide shaft **21** abutted thereagainst, and the fixing member **40** rotates in the clockwise direction in an interlocked manner. That is, since the movement of the slide shaft **21** is not regulated by the fixing member **40**, drawbacks such as deformation and damage may be suppressed even if the cover **10** is closed from the state of FIG. **30A**. Moreover, as illustrated in FIG. **30B**, the fixing member **40** rotates in the clockwise direction and the slide shaft **21** moves to a position where the slide shaft **21** and the non-support surface **44** are separated. Thereafter, the fixing member **40** is caused to automatically rotate in the counterclockwise direction to the first state illustrated in FIG. **30A** by an elastic member not shown, such as a spring. Thereby, the user is not required to return the fixing member **40** to the first state illustrated in FIG. **30A**, and the cover **10** can be retained at the first angle without fail when the cover **10** is opened for the next time from the closed state.

(142) As described above, according to the image forming apparatus **1** of the present embodiment, the fixing member **40** in the second state automatically moves to the first state, such that the user is required to simply close the cover **10**, according to which the operability may be enhanced.

(143) According to the image forming apparatus **1** of the above-mentioned embodiment, a case in which the guide portion **330** is linear has been described, but the present technique is not limited thereto. For example, as illustrated in FIG. **31**, similar to the first to third embodiments, the first path **331** and the second path **332** may adopt stepped shapes. In this case, when the slide shaft **21** is positioned at an end portion **331a** of the first path **331**, the support arm **20** is in a balanced state by the rotation shaft **22** and the slide shaft **21** positioned at the end portion **331a**. The cover **10** is retained at the first angle by the cover rotation shaft **18** and the slide shaft **21** positioned at the end portion **331a** being balanced. According to the configuration illustrated in FIG. **31**, when the cover **10** is opened to the first angle from the state closed with respect to the image forming apparatus **1**, the fixing member **40** closes a connecting portion between the first path **331** and the second path **332** such that the slide shaft **21** is prevented from entering the second path **332**. That is, the slide shaft **21** is guided toward the end portion **331a** by a guide surface **43b** of the fixing member **40**. Thereby, in a state where the cover **10** is retained at the first angle, the slide shaft **21** maintains a state where load is not applied to the fixing member **40**. By adopting this configuration, it becomes possible to reduce the stiffness and strength of the fixing member **40**, that is, it becomes possible to use a small member formed of a lightweight material, such that the image forming apparatus **1** can be downsized with reduced weight.

(144) According to the image forming apparatus **1** of the above-mentioned embodiment, after reaching the second state and the slide shaft **21** has passed the fixing member **40** toward the end portion of the guide portion **330** illustrated in FIG. **29B**, the fixing member **40** is caused to return automatically to the first state illustrated in FIG. **29A**. However, the present technique is not limited thereto, and for example, similar to the first to third embodiments, the fixing member **40** may be caused to maintain the second state after the slide shaft **21** has passed the fixing member **40** toward the end portion of the guide portion **330** illustrated in FIG. **29B**. In that case, when closing the cover **10** from the second angle to the first angle, the slide shaft **21** moves within the guide portion **330** and abuts against a returning surface **42** of the fixing member **40**. Further, by the fixing member **40** rotating in the clockwise direction in an interlocked manner with the movement of the slide shaft **21**, the fixing member **40** is transited to the first state illustrated in FIG. **29A**. Then, when the user returns the rotation stop portion not shown, the fixing member **40** will no longer rotate in the counterclockwise direction from the first state illustrated in FIG. **29A**, and the cover **10** may be retained at the first angle. Of course, by closing the cover **10** further from this state toward the image forming apparatus **1**, the cover **10** may be closed completely. However, even according to this configuration, the rotation stop portion not shown may be returned to the clockwise direction due to external forces applied by an operator or by vibration and collision, before the operation of closing the cover **10** from the second angle to the first angle as illustrated in FIG. **30A**

is performed. In that case, according to the present embodiment, when the slide shaft **21** moves from the state of FIG. **30A** toward the closing direction of the cover **10**, the non-support surface **44** is pressed by the slide shaft **21** in contact therewith, and the fixing member **40** rotates in the clockwise direction in an interlocked manner, such that the cover **10** may be closed.

Fifth Embodiment

(145) Next, a fifth embodiment of the present invention will be described in detail with reference to FIG. **32**. The configuration of the present embodiment differs from the fourth embodiment in that it has a function to retreat from the first state by having a tip of the fixing member rotate. Other configurations are similar to the fourth embodiment, such that the same reference numbers are assigned to corresponding components, and detailed descriptions thereof are omitted.

(146) FIG. **32A** is an enlarged view illustrating from a front side of the apparatus a movement of the slide shaft **21** in a state where the cover **10** according to the present embodiment is closed from a second angle to a first angle. FIG. **32B** is an enlarged view illustrating from a front side of the apparatus a movement of the slide shaft **21** and the fixing member **40** in a state where the cover **10** is closed from the second angle to the first angle. As illustrated in FIG. **32A**, the movable member **60** is provided on the fixing member **40**. The movable member **60** is supported rotatably about a rotation shaft **61** with respect to the fixing member **40**, and rotation thereof is restricted by a rotation stopper not shown so as not to rotate in the counterclockwise direction with respect to the fixing member **40** from the state of FIG. **32A**. Thereby, when the slide shaft **21** moves from the state of FIG. **32A** toward the direction closing the cover **10**, the movable member **60** is pressed by the slide shaft **21** in contact therewith, and the movable member **60** is rotated in an interlocked manner in the clockwise direction. That is, since the movement of the slide shaft **21** is not restricted by the movable member **60**, the cover **10** may be closed from the state of FIG. **32A**.

(147) Furthermore, as illustrated in FIG. **32B**, the movable member **60** rotates in the clockwise direction, and the slide shaft **21** is moved to a position where the slide shaft **21** and the movable member **60** are separated. Thereafter, the movable member **60** is rotated in the counterclockwise direction with respect to the fixing member **40** automatically to the position illustrated in FIG. **32A** by an elastic member not shown, such as a spring. Thereby, the user is not required to return the movable member **60** to the state of FIG. **32A**, and the cover **10** can be retained at the first angle without fail when the cover **10** is opened for the next time from the closed state.

(148) According to the present invention, it becomes possible to simplify the operability when opening the cover in two stages.

OTHER EMBODIMENTS

(149) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(150) This application claims the benefit of Japanese Patent Application No. 2022-177474, filed Nov. 4, 2022 which is hereby incorporated by reference herein in its entirety.

Claims

1. An image forming apparatus comprising: an image forming unit configured to form an image on a sheet; a casing accommodating the image forming unit; a door configured to be opened and closed with respect to the casing, and configured to be moved to a first opened position and a second opened position, the first opened position being a position where the door is inclined by a first angle with respect to a closed position, where the casing is closed and a sheet conveyance path is formed, to expose the sheet conveyance path, the second opened position being a position where the door is inclined by a second angle that is greater than the first angle with respect to the closed position to expose the sheet conveyance path; a connecting member configured to be interlocked

with an opening and closing operation of the door, and connected to the casing and the door; a guide member provided on the door, the guide member including a first guide configured to guide the connecting member in a case that the door is moved between the closed position and the first opened position, and a second guide branched from the first guide and configured to guide the connecting member in a case that the door is moved to the second opened position, wherein the connecting member is configured to retain the door at the first opened position by engaging an end of the first guide and to retain the door at the second opened position by engaging an end of the second guide; and an operating portion configured to move, through an operation by a user and in a state where the door is retained at the first opened position by the connecting member engaging the end of the first guide, from a regulating position where the connecting member is regulated from moving from the first guide to the second guide, to an allowing position where the connecting member is allowed to move from the first guide to the second guide, wherein, in a state where the operating portion is at the allowing position, the door is configured to be retained at the second opened position by movement of the connecting member from the first guide to the second guide.

2. The image forming apparatus according to claim 1, wherein the connecting member is configured to be switched between a first state in which the door is rotatable from the closed position to the first opened position, and a second state in which the door is rotatable at least from the first opened position to the second opened position, and wherein the operating portion is configured to switch the connecting member between the first state and the second state.

3. The image forming apparatus according to claim 2, further comprising: a regulation portion configured to be interlocked with the operating portion, wherein the connecting member includes a guided portion configured to be guided by the guide member, wherein the regulation portion is configured to regulate a movement of the guided portion, wherein, in a state where the guided portion is engaged with the first guide and the operating portion is at the regulating position, the regulation portion is configured to regulate the guided portion from entering to the second guide, by which the connecting member is in the first state, and wherein, in the state where the operating portion is at the allowing position, the connecting member is in the second state by the first guide and the second guide being communicated.

4. The image forming apparatus according to claim 3, wherein the connecting member is configured to be switched from the first state to the second state by an operation of the operating portion being moved from the regulating position to the allowing position, and wherein the connecting member is configured to be switched from the second state to the first state by rotating the door from the second opened position to the closed position.

5. The image forming apparatus according to claim 2, further comprising: a regulation portion configured to be interlocked with the operating portion, wherein the connecting member includes a guided portion configured to be guided by the guide member, wherein the regulation portion includes a contact portion configured to intersect a trajectory of the guided portion in the first guide in a state where the connecting member is in the second state, and wherein, while the door moves from the second opened position to the closed position, the guided portion comes in contact with the contact portion to move the regulation portion from the allowing position to the regulating position.

6. The image forming apparatus according to claim 2, further comprising: a regulation portion configured to be interlocked with the operating portion, and a retaining portion configured to retain the regulation portion such that the connecting member is maintained in the first state, wherein the regulation portion is released from being retained by the retaining portion by an operation of the operating portion being moved from the regulating position to the allowing position.

7. The image forming apparatus according to claim 3, wherein the guide member includes a receiving portion configured to receive a load of the door acting on the guided portion in a state where the door is positioned at the first opened position.

8. The image forming apparatus according to claim 1, wherein the operating portion is configured

to move to the regulating position and the allowing position.

9. The image forming apparatus according to claim 1, wherein the operating portion is disposed at a portion of the door other than a portion forming an external wall of the casing in the closed position.

10. The image forming apparatus according to claim 1, wherein the door forms a side wall portion of the casing in the closed position, and is configured to rotate about a rotation shaft in a horizontal direction at a lower part of the door.

11. An image forming apparatus comprising: an image forming unit configured to form an image on a sheet; a casing accommodating the image forming unit; a door configured to be opened and closed with respect to the casing, and configured to be moved to a first opened position and a second opened position, the first opened position being a position where the door is inclined by a first angle with respect to a closed position, where the casing is closed and a sheet conveyance path is formed, to expose the sheet conveyance path, the second opened position being a position where the door is inclined by a second angle that is greater than the first angle with respect to the closed position to expose the sheet conveyance path; a connecting member connected to the casing and the door; a guide member provided on the door, the guide member including a first guide configured to guide a guided portion of the connecting member in a case that the door is moved between the closed position and the first opened position, and a second guide branched from the first guide and configured to guide the guided portion in a case that the door is moved to the second opened position, wherein the connecting member is configured to retain the door at the first opened position by engagement of the guided portion of the connecting member and an end of the first guide and to retain the door at the second opened position by engagement of the guided portion of the connecting member and an end of the second guide; and a regulating portion configured to move, in a state where the door is retained at the first opened position by the guided portion of the connecting member engaging the end of the first guide, from a regulating position where the guided portion is regulated from moving from the first guide to the second guide by contact with the regulating portion, to an allowing position where the guided portion is allowed to move from the first guide to the second guide, wherein, in a case where the door is opened in a state where the regulating portion is at the regulating position, the guided portion is regulated from moving from the first guide to the second guide by the regulating portion, and wherein, in a case where the door is opened in a state where the regulating portion is at the allowing position, the guided portion moves from the first guide to the second guide to allow the door to be at the second opened position.

12. The image forming apparatus according to claim 11, further comprising: an operating portion configured to be operated by a user to move the regulating portion.

13. The image forming apparatus according to claim 11, wherein the regulating portion is configured to move independently from the connecting member.

14. The image forming apparatus according to claim 11, wherein the regulating portion includes a contact portion configured to intersect a trajectory of the guided portion in the first guide in a state where the door is at the second opened position, and wherein, while the door moves from the second opened position to the closed position, the guided portion comes in contact with the contact portion to move the regulating portion from the allowing position to the regulating position.
